

# Understanding User Privacy Concerns of Shared Smart TVs

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As smart TVs gain popularity, they introduce significant privacy and security concerns due to their extensive data collection and multi-user contexts. This paper investigates user perceptions of privacy concerns regarding both service providers and the multi-user use case in the context of smart TVs. Through in-depth interviews with 22 smart TV users, we found that participants expressed uncertainty about the data collection practices of smart TVs and a desire for clearer communication of such practices. Participants reported discomfort with how their personal information is handled through their smart TVs but felt forced to accept it due to the lack of ability to opt out. Our study also highlights varied privacy concerns when smart TVs are shared in public versus private settings. While participants expressed significantly less concern when sharing smart TVs with acquaintances in private settings, concerns were more prevalent in public settings like hotels and Airbnbs. Based on the findings, we provide recommendations for designers, policymakers, and researchers to improve privacy protection and user experience around smart TVs.

CCS Concepts: • **Security and privacy** → **Privacy protections; Social aspects of security and privacy**; • **Human-centered computing** → **Empirical studies in HCI**.

Additional Key Words and Phrases: privacy, smart TV, Internet of Things

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## 1 Introduction

Smart TVs are advanced television systems that enable users to access streaming services. Unlike traditional televisions, which broadcast predetermined content, smart TVs allow users to request and view content on demand. With smart home Internet of Things (IoT) devices gaining widespread adoption, smart TVs have emerged as one of the most popular home electronics, with the number of households owning smart TVs constantly growing in recent years and is predicted to reach around 117 million in 2025 in the United States alone [4]. Despite the convenience and enhanced entertainment experiences they offer users, smart TVs raise significant data privacy and security concerns, due to their feature of being “smart” and commonly “shared” in use.

The “smartness” of smart TVs indicates their capacity to serve as a channel for extensive data collection by service providers. These service providers can include the manufacturer of smart TV hardware or operating system (OS), as well as third-party apps installed by users. Despite the lack of studies that solely focus on smart TVs, researchers have broadly studied user’s privacy concerns of general IoT devices regarding their data collection practice [13, 33, 36, 50, 64, 72], yet privacy norms

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are highly correlated to the context of data flow, such as what data is collected, who collects the data, and how it is processed [52]. In addition, several studies that investigated user perceptions of privacy and security on specific IoT devices (e.g., smart speakers and smart locks [25, 33, 45]) have shown nuanced perception differences compared to general IoT devices because of the significant variance in data collection pipelines across different devices. User privacy concerns specifically associated with smart TV service providers (referred to as *provider privacy* hereafter), therefore, cannot be accurately represented by results from studies on general IoT devices and need separate attention.

Smart TVs are also commonly “shared” among multiple users for long-duration and in-depth interactions during social activities. This usage pattern suggests the collection of data from and shared access for not only smart TV owners and direct users, but also other residents and visitors in the household, and even people passing by the residence. In this paper, we use the term “bystander” to refer to the latter group of people. In the case of shared smart home, tensions between IoT device owners/primary users – who hold primary controls over devices – and bystanders on privacy setting preferences [6, 8, 14, 71] and social boundaries [8, 21, 45, 73] have been widely identified (referred to as *social privacy* hereafter). While smart home IoT devices have been extensively studied from this perspective, perceptions specifically regarding shared smart TVs have been overlooked. Moreover, smart TVs can be shared in public settings where the owners of the smart TVs are not acquaintances of the users, which is rarely explored in studies of general IoT devices.

With the above research gaps in mind, we propose the following research questions:

- RQ1: What privacy concerns do users have regarding the collection of their data through smart TVs?
- RQ2: How satisfied are users with the current privacy management systems on smart TVs?
- RQ3: What privacy concerns do users have when sharing their smart TVs with others?
- RQ4: What privacy concerns do users have when using smart TVs owned by others?

To answer the research questions, we interviewed 22 participants in the United States to investigate their privacy concerns related to using smart TVs. We guided the interviews with scripts developed to i) explore user perception of potential privacy violations by data practice of service providers (provider privacy); and ii) detangle the complex social dynamics that arise when smart TVs are shared among multiple users, both in private settings like households and public settings like hotels and Airbnbs (social privacy).

For provider privacy (RQ1, RQ2), we found that despite the appreciation for the accessibility and variety of content offered by smart TVs, participants expressed major privacy concerns about the specific types of information collected and how it was used. Although participants accepted trading their data for convenient services, they were usually forced to do so and perceived a lack of choice to not provide their data. In addition, participants generally found privacy practices on smart TVs to be poorly communicated and difficult to understand, and advocated for clearer explanations and more accessible privacy settings.

For social privacy (RQ3, RQ4), participants perceived sharing smart TVs with acquaintances as a relatively low-risk activity. When sharing with acquainted bystanders in their homes, participants were aware of the information bystanders could access, but often expressed limited concern due to the perceived nonsensitive nature of the data and trust in acquainted bystanders. However, some participants felt social discomfort or had safety concerns, leading to measures such as clearing viewing history regularly or setting parental controls for minors. When interacting with other people’s smart TVs, participants expressed little concern when the TV belonged to acquaintances. In public spaces, such as hotels or Airbnbs, concerns were more prevalent, particularly regarding the use of accounts and the potential data collection.

Our study makes important contributions to the understanding of privacy concerns related to smart TVs based on empirical evidence collected through interviews. We believe our focus on a single IoT device and our qualitative approach capture user opinion and attitudes more precisely and in detail than studies with a broader scope which cover multiple IoT devices [34, 36, 50, 75] and the quantitative approach [6, 8, 14, 27, 28, 40, 50] more commonly applied. Complementing studies that focus on data collected through sensors, e.g. visual, audio, or environmental data [13, 25, 33, 36, 64, 75], we focus on data related to user preference, especially the types of content they consume. Our study highlights the lack of clear communication or control over personal data collected by smart TVs. Additionally, we examine the implications of shared smart TV use, particularly the dynamics between owners and bystanders in both private and public settings, which we found is more complex than that of sharing other IoT devices because of the unique features and functionalities of smart TVs. By addressing both provider and social privacy issues, our work fills a significant gap in the literature on smart TV privacy, offering concrete recommendations for product designers, policymakers, and researchers. We hope our work can help enhance privacy protections and improve the overall user experience of smart TVs.

## 2 Related Works

### 2.1 Smart TV Security Risks

Smart TVs are known to have significant security vulnerabilities as reviewed in [67]. Researchers have highlighted that the architecture of Smart TVs can be susceptible to malware and security breaches through various channels, including email, cameras, Internet connectivity, and media players [57, 58]. More specifically, it is possible to gain root access to smart TV systems and exploit built-in cameras or microphones [38, 46, 47]. Remote controls, the default and most commonly used medium for operating smart TVs, introduce security risks as well. For instance, Kannan et al. successfully recovered information entered via remote controls by analyzing the acoustic signals emitted by Smart TVs [29]. Zhang et al. developed a multi-channel remote control mimicry attack, which allows attackers to gain control of victims' smart TVs [74].

In addition to system robustness, researchers have identified security risks of third-party apps available on smart TVs. For example, Niemietz et al. examined the network traffic of popular apps and found that some apps transmit login data over unencrypted channels [51]. Similarly, Mohajeri et al. discovered that major service providers, such as Roku and Amazon Fire TV, send unique identifiers to known trackers sometimes over unencrypted connections [48].

### 2.2 Smart Home IoT Privacy

Usable Security and HCI researchers have investigated people's awareness, attitudes, and actions regarding personal privacy and system security in smart homes. In this section, we summarize two threads of prior works in this domain: end-user perceptions of data privacy and security of smart home IoT, and privacy-related social tensions of shared IoT devices.

**2.2.1 End-User Perceptions on Data Privacy and Security.** Prior works have widely reported user concerns on smart home IoT about personal privacy such as unclear data flow, and security incidents such as data breaches and hacking [13, 33, 36, 50, 64, 72]. Privacy and security could play an important role when people purchase and interact with smart home IoT devices [17, 67].

However, researchers also found people do not often pay significant attention to privacy and security, weighing them less than other factors when talking about smart home IoT. The misalignment between users' high privacy awareness and limited privacy concerns with poor protective actions when using IoT devices has been commonly acknowledged [27, 50, 53, 64, 72]. Meanwhile, users are likely to trade their privacy for the benefits brought by smart home IoT [24, 49, 64, 75].

Users' low concern over privacy and security with smart home IoT leads to their limited use of control mechanisms and high reliance on ad-hoc strategies to protect privacy, as reported in prior works [28, 53, 72].

Despite people's overall low privacy concerns of about smart home IoT, researchers found user perceptions on IoT devices to highly correlate with the context of data collection and processing [1, 9]. Specifically, prior works revealed aggregated privacy and security concerns using smart home IoT in certain situations – with certain data collected, with certain entities collecting the data, and with certain end-users. For instance, studies have shown that users were most concerned about physical information like visual and audio [13, 36, 64], while they were unaware of risks from non-video/audio collected devices [75], and were likely to consent to data practice that enhances user experience [49]. Moreover, user concerns about different data collectors are affected by their trust in those entities [70, 75]. Researchers have also studied users' perceptions of specific types of smart home IoT devices. For example, Lau et al. studied people's privacy concerns around smart speakers [33], and Hazazi et al. investigated end users' privacy and security perception of smart locks, considering their tight connection to users' physical safety [25].

Unique features of smart TVs could introduce security risks, as reviewed in Section 2.1. Furthermore, studies have revealed that functionalities such as Automatic Content Recognition (ACR) [7] and digital activity logs [30] can raise significant privacy concerns by enabling the tracking of user behavior. Researchers have therefore investigated people's attitudes toward security incidents in smart TVs [22] and privacy in sensor-based smart TVs [39] using large-scale surveys. However, end-users' nuanced perceptions of data, privacy, and security in smart TVs have remained unexplored. Through an in-depth interview study, our work contributes to the body of knowledge by investigating users' mental models of the data pipeline within smart TV systems, revealing their privacy perceptions and privacy management behaviors.

**2.2.2 Privacy and Social Tensions on Shared Devices.** The interaction with smart home IoT devices, including smart TVs [31, 44], often involves multiple individuals in a household, thereby introducing privacy concerns and social tensions. Prior works have revealed smart home bystanders' uncertainties and concerns about data collected by smart home devices, as well as a strong need for privacy control [5, 6, 41, 71]. As bystanders usually have limited data and privacy control over IoT devices, they sometimes negotiate with owners/primary users around privacy settings [6, 14, 21, 34, 68]. Despite the growing body of research concerning bystander privacy, researchers have underlined a critical and common gap in privacy and security expectations and preferences between owners/primary users and bystanders [6, 8, 14, 71].

Besides the misalignment of privacy preferences, researchers have also noticed conflicts in device settings, controls, access, and usage between owners/primary users and bystanders, which creates social tensions [8, 21, 45, 73]. These tensions could be more complex in shared households, for example, between guests and hosts in Airbnbs [40]. The potential misuse and abuse of IoT devices by owners/primary users can lead to interpersonal abuse [63]. Users are concerned about the potential use of smart home IoT for spying on residents, visitors, and neighbors [23, 72], enhancing intimate partner violence (IPV) in adversarial households [35, 62], and monitoring domestic workers [10, 11], all of which could severely violate social boundaries and compromise personal privacy rights. Additionally, owners/primary users are concerned about bystanders violating their personal privacy by obtaining their private information or processing malicious actions through their accounts [14, 15, 33, 41]. These concerns on shared IoT devices can be heightened in smart TVs due to their wide presence in public spaces such as Airbnbs and hotels [26, 40, 42].

The potentiality of privacy and social tensions among individuals around shared smart TVs could be different from that around other smart home devices. Our work, therefore, attempts to fill in

the gap and looks into people's perceptions of privacy, social boundaries, and tensions of sharing smart TVs.

### 3 Methodology

To answer the research questions, we conducted semi-structured interviews with 22 smart TV users. This study was approved by our institution's Institutional Review Board (IRB), and all participants provided informed consent before their interviews. All surveys and interview scripts can be found in Appendix B.

#### 3.1 Recruitment and Participants

Our participant recruitment took place between January and July 2024. Participants were recruited through Prolific<sup>1</sup>, a crowd-sourcing platform for research studies, and within the university. To avoid getting mostly privacy-preserving participants, we described our study as "understanding user experience of using smart TVs" and did not mention anything about privacy and security. Interested individuals first completed a screening survey, which collected general information about their smart TV usage and demographic details. Participants who completed the screening survey received a compensation of \$2. We then manually selected participants who (i) use smart TVs at least weekly if they own smart TVs and monthly if they do not, (ii) interact with smart TVs in diverse scenarios, and (iii) use smart TVs for a wide range of purposes. We did not avoid smart TV users who did not own smart TVs in their homes since they could provide valuable insights into using smart TVs in others' residences or public spaces. Selected participants were then invited to participate in a semi-structured interview. Recruitment ended after no significantly new use cases or privacy concerns related to smart TV usage emerged, indicating that data saturation had been reached.

We finally recruited 22 participating smart TV users, consisting of nine females, twelve males, and one non-binary ranging from 18 to 55 years old or older. Four of the participants (P5, P7, P10, P11) did not own any smart TVs but had enriched experience using smart TVs in different places, while the rest 18 participants were all smart TV owners. Table 1 provides detailed information about the participants and their smart TVs. Participants reported a diverse range of interactions with smart TVs. Some described watching smart TV as a way to unwind after work, while others, particularly retired individuals, enjoyed watching smart TVs with their grandchildren. Some participants also shared their smart TVs primarily with partners, whereas others frequently used them in social settings with friends. Table 2 in Appendix A presents additional demographic information to further characterize the participant population.

#### 3.2 Interview Procedure

Participants who wish to proceed completed a pre-interview survey that asked for more detailed information about their smart TV usage, such as where and with whom they watch smart TVs. At the end of the survey, they were redirected to a Calendly<sup>2</sup> link to schedule their interview, and were instructed to use their Prolific ID and email address to remain anonymous. All interviews were conducted via Zoom and audio-recorded. Two researchers on the team attended each interview, with one serving as the primary interviewer and the other acting as a note-taker while ensuring that all questions of interest are addressed. Participants who completed the interview were compensated \$25.

<sup>1</sup><https://www.prolific.com>

<sup>2</sup><https://calendly.com>

The interviews began with questions about participants' general understanding and use cases of smart TVs. We then explored their privacy concerns related to smart TV usage, structuring the interview around two key themes:

**Provider Privacy** We explored participants' mental model of the data pipeline of smart TVs, namely the types of information they believed were collected through smart TVs, which providers collected this information, and for what purposes. Furthermore, we investigated their privacy concerns about the data pipeline and inquired about their satisfaction level regarding how well providers inform them about the pipeline.

**Social Privacy** We examined how participants shared their smart TVs and accounts with acquaintances and discussed their privacy-related concerns regarding these practices. Additionally, we explored participants' use of smart TVs in acquaintances' residencies and public spaces such as hotels and Airbnbs, and their perceived privacy risks.

To prevent leading participants to imagine non-existent privacy concerns, we deliberately refrained from mentioning any privacy- or security-related terms until the final part of the interview, unless participants brought up related topics by themselves. We also framed the questions in a neutral manner, and were able to observe that many of our participants reported no or limited privacy concerns to some aspects of smart TVs.

The interviews concluded with general questions on privacy and security. Participants were asked if they were aware of any security attacks on smart TVs and their overall concerns about technology. We also gathered self-rated privacy awareness levels using a scale of low, medium, and high but found no correlation between them and participants' perceptions.

### 3.3 Data Analysis

To analyze the interviews, we obtained transcriptions of the recordings in plain English text. The transcriptions were then imported to MAXQDA<sup>3</sup> to perform open coding. We developed the codebook based on the interview scripts first and two researchers on the team performed a round of trial coding on the first five interview scripts. After the two researchers completed the coding individually, they met to resolve inconsistencies. With the utilization of thematic analysis [12], codes for topics frequently mentioned by participants were then added and excess codes were combined or dropped. The two researchers then recoded the first five interview scripts and coded another five interview scripts to test the functionality of the codebook and discussed inconsistencies again to make minor adjustments. The final codebook consists of 22 general codes organized into 5 categories, namely "General usage", "Data collection and processing", "Private space sharing", "Public space sharing", and "Privacy and security", with 11 of the general codes further divided into 35 sub-codes in total. For example, under the "Data collection and processing" category, we have the general code "Private info definition", which does not have further sub-codes, while the general code "Manufacture" has three sub-codes, namely "Manufacture collect", "Manufacture process", and "Manufacture comfort level." The complete codebook can be found in Appendix C. Every transcript was coded using the final codebook in the end. Inconsistencies between two researchers were discussed in regular meetings to resolve issues such as whether text should be labeled with multiple codes. Since we performed open coding for thematic analysis, inter-rater reliability score is not applicable and therefore not calculated [43]. Our synthesized findings are presented in section 4 and section 5.

<sup>3</sup><https://www.maxqda.com>

| Participant | Gender                    | Age                   | Number of TVs owned | TV Brands owned                                                                                      |
|-------------|---------------------------|-----------------------|---------------------|------------------------------------------------------------------------------------------------------|
| P1          | Male                      | 25-34 years old       | 1                   | LG TV - WebOS                                                                                        |
| P2          | Non-binary / third gender | 25-34 years old       | 3 or more           | Amazon Fire TV - Fire OS, VIZIO - Smartcast                                                          |
| P3          | Male                      | 35-44 years old       | 3 or more           | ROKU / AOC - Roku OS                                                                                 |
| P4          | Female                    | 55 years old or older | 2                   | ROKU / AOC - Roku OS                                                                                 |
| P5          | Female                    | 25-34 years old       | 0                   | -                                                                                                    |
| P6          | Male                      | 25-34 years old       | 1                   | ROKU / AOC - Roku OS                                                                                 |
| P7          | Male                      | 25-34 years old       | 0                   | -                                                                                                    |
| P8          | Female                    | 25-34 years old       | 1                   | Hisense - VIDAA                                                                                      |
| P9          | Male                      | 25-34 years old       | 1                   | LG TV - WebOS                                                                                        |
| P10         | Male                      | 18-24 years old       | 0                   | -                                                                                                    |
| P11         | Male                      | 25-34 years old       | 0                   | -                                                                                                    |
| P12         | Male                      | 25-34 years old       | 1                   | Amazon Fire TV - Fire OS, VIZIO - Smartcast                                                          |
| P13         | Male                      | 35-44 years old       | 2                   | Amazon Fire TV - Fire OS, ROKU / AOC - Roku OS                                                       |
| P14         | Male                      | 55 years old or older | 1                   | Sony / TCL / Xiaomi / Skyworth / Sharp / Philips / Chromecast / TiVo / Hisense - Google / Android TV |
| P15         | Female                    | 55 years old or older | 2                   | Hisense - VIDAA                                                                                      |
| P16         | Female                    | 45-54 years old       | 3 or more           | ROKU / AOC - Roku OS, Panasonic - Firefox OS                                                         |
| P17         | Male                      | 25-34 years old       | 1                   | Sony / TCL / Xiaomi / Skyworth / Sharp / Philips / Chromecast / TiVo / Hisense - Google / Android TV |
| P18         | Male                      | 35-44 years old       | 2                   | ROKU / AOC - Roku OS                                                                                 |
| P19         | Female                    | 45-54 years old       | 3 or more           | ROKU / AOC - Roku OS                                                                                 |
| P20         | Female                    | 45-54 years old       | 3 or more           | Samsung TV - TIZEN OS, LG TV - WebOS                                                                 |
| P21         | Female                    | 25-34 years old       | 1                   | Amazon Fire TV - Fire OS                                                                             |
| P22         | Female                    | 25-34 years old       | 2                   | ROKU / AOC - Roku OS                                                                                 |

Table 1. Participants demographics information

### 3.4 Limitations

While we strive to make our findings as representative as possible, our study has limitations inherent to the interview method and the types of participants we were able to recruit. All participants resided in the United States and were influenced by American social perspectives. Additionally, our participant pool is slightly skewed towards males (12 males, nine females, and one non-binary participant), which may impact our results due to potential differences in privacy awareness between genders [32, 65]. Future works may explore different populations based on cultural background and levels of technical expertise.

Additionally, we were limited to the types of smart TV users we could recruit. Despite our efforts to include participants who do not own smart TVs, non-owners remain underrepresented in our study. This underrepresentation may result in fewer reports of dissatisfaction, as individuals with

strong negative attitudes toward smart TVs may be less likely to purchase them. We also did not explicitly sample participants based on household types, which may have limited our exploration of the variety of relationships participants have with their co-watchers. Lastly, there is a lack of participants who had smart TVs with cameras or microphones, and user perception of related functionalities therefore remains unclear.

## 4 Provider Privacy

In this section, we present our findings on how participants considered their privacy regarding the complex data practices across different service providers of smart TVs. We begin by presenting participants' general understanding and background of using smart TVs, followed by an examination of their understanding of data practice and privacy concerns. Finally, we demonstrate participants' opinions on provider privacy practices, including the communication of these practices, and the setting options available on their smart TVs.

### 4.1 Conceptions and Usage of Smart TVs

When describing a smart TV, most of the participants regarded it as an Internet-connected, streaming service-integrated TV. Participants found smart TVs to have similar functionalities as other smart devices, such as smartphones and tablets, but with a more specific focus on content consumption.

Among the four participants who did not own a smart TV, two cited privacy concerns as their primary reason for not purchasing one. Other reasons included a perceived lack of necessity. Meanwhile, owner participants identified features and price as the primary factors they considered when making their purchases, and did not list privacy and security features as their major considerations.

All participants identified watching online content as their primary use of smart TVs. Other usages spanned casting other devices and gaming monitors. Eleven of eighteen participants who owned smart TVs used remote controls exclusively to operate their smart TVs, appreciating their reliability and convenience. Ten participants expressed concerns about the potential exchange of personal data when using a smartphone or smart speaker to control their smart TVs. Three participants acknowledged the possibility of data exchange, but did not consider it significant because their information is already "out there."

### 4.2 Privacy Concerns

All participants were able to list some types of information they believed their smart TV was collecting. While the information mentioned was considered personal or private, six participants noted that not all of it was sensitive or a source of privacy concern. Participants were able to identify their smart TV manufacturers and the third-party apps they use on their devices as the primary data collectors directly gathering information from their devices. Additionally, two participants mentioned their Internet service providers as data collectors, although the discussion was less directly relevant to smart TVs and is therefore not presented here.

In the following sections, we focus on participants' perceptions of smart TV manufacturers and third-party apps specifically.

#### 4.2.1 Hardware/OS Manufacturers.

*Understanding of data collection.* Some participants had divided awareness of the data practice of their smart TV manufacturers. One participant admitted that they could not think of any information that would be useful for the manufacturer to collect, while three others believed that manufacturers would collect as much data as possible. As P10 commented:

*Oh, everything. They have control on the hardware level. They can build whatever they want. It is just how much can they get away with. (P10)*

Most participants, however, were uncertain about the specific types of data being collected. The most frequently mentioned information included app usage statistics and demographic or profile details, cited by nine and eight participants, respectively. P6 shared an anecdote with us:

*They also definitely know my name [and] email address, because I had to create an account to even use the TV. They know the kind of consoles I have attached to my TV because I've seen ads for games that came up in my email inbox only after I had used this TV and connected my consoles to it, that I hadn't gotten before. (P6)*

Additionally, participants speculated that manufacturers might collect device statistics useful for product improvement purposes. Three participants suggested that their smart TVs might record screen activity, potentially capturing information about the apps they used, the content they consumed, and even payment details entered.

As to the purpose of data collection, 16 participants believed that television manufacturers would either sell their data to third-party companies or use it to deliver targeted advertisements to other businesses. Additionally, 11 participants indicated that the data might be used by manufacturers to enhance product development or to forecast future sales.

*Comfort level.* Participants generally exhibited low levels of comfort regarding the collection and use of their information by smart TV manufacturers. Only two participants felt completely comfortable with the data practices of these companies. In contrast, 11 participants emphasized their negative attitudes towards these data practices. P6 questioned the fairness of having to provide manufacturers additional information to use the device and highlighted concerns not only with privacy protection but also with users' rights over their own data:

*I should not have my information sold just to use my TV. It does not need any of these pieces of information if I just want to plug in an external box and watch on that box or if I want to just connect a console and play on the console. (P6)*

Both P4 and P9 expressed regret about not managing their privacy and information better when purchasing and setting up their smart TVs:

*Now, if it were a situation where I knew back then when I bought these devices what I know now, I would have been more persistent about not giving so much detail, but I didn't foresee that it would become this convoluted. (P4)*

*I would've rather just not gotten the TV and it would've saved me some trouble [...] Adding this one more company whose TV I'm using, just adds one more person or one more organization who's going to be taking information and trying to benefit out of it. (P9)*

Nine participants admitted their discomfort but acknowledged that they had accepted how things work or were willing to make the trade-off for better service for now. Five participants expressed confusion about how to protect their privacy in this situation. Specifically, P8 complained about the difficulty involved in preventing his information from being collected:

*I have considered [the issues with TV manufacturer collecting my information]. But for the time being, the investment for return didn't seem quite right and also the effort to make sure that the TV is not collecting anything. (P8)*

#### 4.2.2 Third-Party Apps.

*Understanding of data collection.* Participants demonstrated a deeper understanding of the types of data that third-party apps on their smart TVs might collect. 15 participants identified their watch history as being collected, while 12 mentioned that demographic information or details from their

account profiles were likely gathered. Additionally, five participants noted that their payment information might be collected due to its use for subscription payments or movie rentals.

When asked about reasons that third-party apps collected the information mentioned above, participants uniformly pointed to revenue generation, while describing various methods by which service providers could achieve this. The most frequently mentioned method was providing targeted advertisements, noted by 15 participants, closely followed by making personalized recommendations, which was mentioned by 13 participants. Five participants also speculated that service providers might sell their information to other companies.

*Comfort level.* Again, participants exhibited generally low levels of comfort regarding the collection and use of their information by third-party app service providers. Only four participants reported feeling comfortable with this practice. In contrast, 11 participants expressed strong reluctance to give their data to third-party apps. Some of these participants, though, indicated that they had become accustomed to their discomfort, while others believed that their smart TVs did not significantly contribute to data collection, as their information was already being gathered from other devices.

Seven participants were strongly opposed to the analysis and targeting practice, especially advertising, processed by third parties. For instance, P11 commented:

*That's kind of the big fear around that, is, as soon as you start collecting data that can be used to make right now predictions about a single individual person [...] all of a sudden now you're getting very targeted, very persuasive advertisements for certain products, or it can be used to even send you, who knows, something as crazy as a political campaign message that might be tailored to getting you to think a certain way. That's really my big fear with those folks. (P11)*

P4 noted that their discomfort stemmed from increased awareness of the practice:

*I thought at first, "Yeah, I think that's cool. My time's not wasted by seeing things that I would never buy anyway." [...] But now that I know what I know now, I see that it's not just simply they want to know the demographics that's watching or to target with ads. [...] And that's a little bit disturbing. (P4)*

For P9, the level of comfort depended on the type of information being collected:

*[If] I looked up my symptoms and it said I have some disease, and so, if that kind of information is being passed around, which is of significant value and importance to me, that is not something that I'm comfortable with. (P9)*

Given that the primary use of smart TVs is content consumption, watch history emerges as a type of data strongly associated with this activity. Participants exhibited mixed attitudes toward the collection and use of their watch history. Eight participants did not perceive their watch history as sensitive information, with P2 finding it particularly useful to have a record of it. Five participants were aware that service providers might utilize their watch history to recommend content, but they either accepted this practice and traded their privacy for convenience or did not care about it. However, despite their agreement to watch history being collected and processed, some participants underlined their persistent discomfort with it. P16 expressed a desire for greater control:

*I don't have anything that I feel like I'm trying to hide or I might be ashamed or embarrassed or potentially not feel comfortable with. But I also wish I had the choice to also just say, "No, I'd rather you not do this." (P16)*

Three participants even wished their watch history had not been collected at all. Among them, P14 mentioned that they used to clear their watch history regularly but admitted that they now do so less frequently because they appreciate the convenience of receiving personalized recommendations.

Seven participants emphasized their lack of knowledge about how to stop third-party apps from processing their data, especially those that were unnecessary. P16 highlighted the lack of options:

*I wish I had some potential choices. I don't care if they use it to tailor the content that they carry. However, I would prefer that I could choose whether they use my information in tracking and tailoring it to their ads or something like that. (P16)*

It seemed that participants could only place hope in service providers not abusing their information:

*I just think that you have to accept it and hope that it isn't used in a way that harms you. (P15)*

### 4.3 Privacy Transparency and Management

Next, we delve into participants' experiences and perceptions regarding managing their data and privacy in smart TVs. We first describe participants' perceptions of the transparency of data and privacy practice and privacy settings on their smart TVs, respectively. Then, we present their expectations for improving the design of privacy-related features on smart TVs.

*Transparency of privacy practices.* Participants expressed uncertainty about where to find information on the data being collected by their smart TVs and how that data was being used, and were dissatisfied with how their devices notify them of these data practices. Some of them pointed to the Terms of Services and Privacy Policy they remembered agreeing to, but also emphasized that they never read them. P2 highlighted the lack of emphasis on important information:

*They don't put it in your face usually necessarily to say like these are important things that you should be paying attention to. (P2)*

Other participants criticized the privacy policies for using "legal jargon" (P11) and being "difficult to understand [and] in deliberately obscure language" (P14). Overall, participants felt that their smart TVs did not effectively inform them about privacy practices, indicating a clear need for straightforward and accessible privacy policies.

*Privacy Settings.* Participants also had limited knowledge about the privacy-related settings on their smart TVs, with fifteen participants mentioning a lack of knowledge about any privacy-related settings available on their smart TVs. Participants shared how they were deterred by the complexity of managing privacy settings.

*I mean now the companies are all very smart. They put the privacy control always in the third layer menu. And when you're in the second layer, none of them has, looks like they have anything to do with privacy at all and you have to really fish through them one by one until you can find the privacy button [...] and I don't like that. (P10)*

*[I]f you go into the settings and try to use stuff that way, it becomes very, very overwhelming, [...] and I don't understand and know what any of it means. [...] And so going into the settings, I have gone in, and looked at it, and been like, "Nope," and just leave. (P20)*

Some participants felt it unnecessary to use privacy settings, regarding it as useless to control their privacy. For example, P6 expressed distrust in the software system:

*I tried to go into my privacy settings and turn off all of the snooping features that I can turn off, but I just don't trust that it's actually fully off. (P6)*

*Expectations and Design Suggestions.* Similar to when discussing smart TV functionalities (Section 4.1), participants drew parallels between their smart TVs and other smart devices, such as smartphones or laptops, when discussing desirable methods for communicating privacy practices and controlling privacy. 11 participants referenced existing systems on these devices or proposed similar designs for smart TVs. The most popular suggestion was a pop-up reminder when information is

being collected, which participants appreciated for its explicitness (P8, P14), simplicity (P1, P20), and ease of access (P22). Additionally, four participants advocated for regular reminders.

*It would be nice to be reminded that, “Hey, if you need to, if you’d like to review it, if you’d like to make changes, this is where you go.” (P18)*

## 5 Social Privacy

In this section, we present findings on how participants viewed shared smart TVs in different socio-technical settings. Our findings on social privacy of smart TVs consist of two parts: owner perceptions of bystanders interacting with their smart TVs, and bystander perceptions of themselves interacting with smart TVs that belong to others.

### 5.1 Owner Perceptions

According to all 15 owners who lived in multi-resident households, every resident, except children who were usually restricted by parental control, had equal access to smart TVs in their household. Moreover, account sharing for different streaming services on smart TVs is common within multi-resident households. All 15 owners who lived in multi-resident households shared accounts with other residents mutually for all or some streaming services, establishing TV-specific shared accounts for every resident. Owners usually had a primary account signed in for a streaming service and hardly ever switched accounts. For streaming services with a multi-profile account function (e.g., Netflix), some owners set multiple profiles in one shared account instead of switching across different accounts.

11 owners we spoke to had experiences sharing their smart TVs with visitors like friends and visiting family members, but their sharing habits differed from sharing among residents. When watching smart TVs with visitors, ten owners would mainly use their own accounts, with the TV controlled by themselves or other residents, attributing the behavior to owners being more familiar with their own TVs. Only three owners let visitors log in to their accounts or control their TVs, respectively. Owners asked for visitors’ accounts for subscriptions that they did not have, or shared accounts for convenience reasons. Additionally, none of the owner participants set up restrictions for visitors to get access to their TVs and accounts.

**5.1.1 Information Disclosure to Bystanders.** When talking about the information disclosed on a smart TV to the physical environment, many owner participants exhibited a fundamental awareness of the information that bystanders can access. Eleven owners found any bystanders, including other residents, visitors, and passers-by, could easily observe their watching information present on the smart TV interfaces, such as recommendation pages, playlists, and watch/search history. In the case of shared account use on smart TVs, eight of the owners thought much of the personal information held by their accounts, such as username, email, and birthday, could be obtained by other residents and visitors if they were controlling TVs. Three owners also underlined sensitive information in their accounts like credit card numbers and passwords. However, they perceived sensitive information as usually hidden and felt there was a low likelihood of bystanders getting it. Specifically, as everyone in the household had equal access to the smart TV, P2 felt that any resident could get their watching and account information on the TV easily, *“if they were using the TV and they wanted to snoop around.”*

*Certain features make information more accessible to bystanders.* Some participants highlighted certain features of the smart TVs that increased the possibility of bystanders getting owners’ information. P11, despite not being an owner, regarded the big screen and limited usability of the

remote control of the smart TV as factors that bring more chances of exposing owner's information to bystanders, especially when typing sensitive information like passwords:

*I think having a 42-inch wide or 60-inch wide or 80-inch wide keyboard that you're clicking buttons very slowly, one at a time, offers a lot greater possibility for somebody who's just a casual observer to be able to record your password. (P11)*

A few owner participants, meanwhile, thought the physical locations of their smart TVs could bring risks of random passers-by shoulder surfing information on the TV interfaces.

*I had to think, because we have actually big side windows, and we actually have a smart TV in the gym too, and that's open. [...] if we have the TV open in the gym, I'm like, "Yeah, people can." (P20)*

P4 held a similar consideration and thought the physical barriers like blinds on their windows could not prevent passers-by from seeing their TV:

*So, when you're outside, just walking down the street, you can tell if the lights are on or if the TV's on even through those shades, because they're blinds, they're not solid. (P4)*

While these features apply to traditional TVs as well, relevant concerns are amplified with smart TVs due to the greater variety of personal information that can be displayed on screens and the more tailored content consumed by users.

**5.1.2 Privacy Concerns Regarding Bystanders.** In contrast to the thorough understanding of information that bystanders could obtain from smart TVs, many owner participants had no privacy concerns about bystanders getting their information through shoulder-surfing or controlling the TV. Moreover, owners usually regarded account sharing on personal smart TVs as convenient and money-saving for paid services, with only a few concerns about account management. We identified two major reasons for owners' limited concern about bystanders intruding on their privacy when sharing TVs and accounts: trust in bystanders, and limitation and insensitivity of information disclosed.

**Trusted Bystanders.** Ten owners we spoke to attributed their lessened concerns to the trust in bystanders within their homes, who, in their cases, were majorly intimate partners, family members, and close friends. We note that this might be due to a biased sample of our owner participants, who were all from non-adversarial households with only friends and family members visiting their homes.

*Even though it could be found, search history and things like that, I don't have as much of a concern about that if it's somebody that I trust who's in my apartment. (P1)*

On the one hand, owners considered all information on their smart TVs as non-private to trusted bystanders, feeling there was "*not a lot to hide*" (P2). On the other hand, they believed that trusted bystanders would not intentionally snoop around on their information when interacting with their smart TVs.

*First, I usually trust the people who visit our home. But also, I think they don't really care, and that's not really something they are motivated to do, taking my information. So, I'm not really worried about it. (P17)*

**limitation and insensitivity of information disclosed.** Six owners considered much of the information bystanders could see on their smart TVs as limited and insensitive.

*None of this information is super personal. It's more like, 'Oh, I don't need friend X to know that I'm currently thinking about Y,' which is popping up in my recommendations. (P6)*

*I think maybe it would just display my email if you went in the settings. So if someone had access to my TV, I don't think I would be that concerned about what they could find.*  
(P21)

**5.1.3 Emergent Tensions around Bystanders.** Regardless of the overall lack of privacy concerns for bystanders, owner participants reported tensions around bystanders accessing or even observing their smart TVs. Some owner participants indicated their discomfort with bystanders' seeing the information, mainly because of a feeling of social boundary intrusion. Additionally, owner participants with children living together also shared their concerns about children accessing their TVs and accounts from a safety perspective.

**Social Awkwardness and Boundary Intrusion.** Six owners, although they did not regard the behavior as malicious, got embarrassed when bystanders saw their watching information. Some owners shared their negative feelings about co-watchers easily accessing their watching information. For example, P2 mentioned their social awkwardness about their friends figuring out what they have watched through the watch history:

*I feel weird if friends see that kind of stuff like, what have I been doing in the last day or last week. That's a little awkward sometimes, depending on what it is that you've been watching essentially.* (P2)

Considering it "breach in my privacy", P9 expressed a feeling of being powerless to hide watching information on smart TVs from co-watchers:

*[My watch history] is just there on the screen, which was never a problem for traditional TVs, it's like whatever is playing on the cable is playing and you have to watch it, but nobody knows what you watched before that moment in time unless you or someone explicitly told them about it [...] sometimes people tend to interfere by going back in history and just saying, "Oh, you saw this."* (P9)

Additionally, a few participants reported mental discomfort about the possible random passers-by shoulder-surfing what they were watching on smart TVs.

*If I'm watching a movie and there's a steamy sex scene and somebody's walking by my house, I wonder how much of that can they see from out in the street and wonder, "Oh my God, is that lady watching porn?"* (P4)

P2, although did not have such concerns because of the private location of their TVs, reflected on their bad feelings as a passer-by who unintentionally saw others' TVs:

*I have looked through people's windows when I've been driving and seen their TVs. And it makes me feel really weird [...] I don't have a whole bunch of windows where my TV is. I would definitely feel more self-conscious if I did.* (P2)

To protect watching information on smart TVs from being accessed by bystanders, two owners reported that they clear search/watch history and other watching footprints occasionally. One owner also closed their blinds to prevent random passers-by from seeing the contents on their smart TVs. However, many owners did nothing even though they had concerns about bystanders intruding into their social boundaries, perceiving it as limited harm and regarding precautions as unnecessary.

**Children Safety.** Four owners with children in their households discussed children's safety around smart TVs in their houses. First, they were concerned about children viewing age-inappropriate content that other residents had watched or appeared on the recommendation list, especially when children used the same account as them:

*Content specifically, thinking about my partner enjoys watching murder mysteries, and he's not at the age where he understands the difference between reality and fantasy. And so he should not be watching anything that has violence currently. (P18)*

Additionally, they also noted the possibility of children accessing sensitive information on the smart TV, such as billing information and passwords.

Regarding these potential threats to children's safety, all owners with children, except P18 whose child was too young to control the TV, set up parental controls on their TVs to restrict children's access.

## 5.2 Using Others' Smart TVs

According to all the participants we spoke to, 19 of them had experience interacting with smart TVs that did not belong to them. Their experiences spanned from sharing smart TVs in acquaintances' residences (i.e., friends' and family members' homes) to using smart TVs in public or shared spaces (e.g., Airbnbs and public lounges). In the following section, we present participants' perceptions and concerns about themselves, as visitors and guests, using other people's smart TVs in different places.

### 5.2.1 Acquaintances' Smart TVs.

*Device Use Behavior.* 10 participants shared their experience of watching smart TVs in acquaintances' residences. Similar to how owner participants shared smart TVs with visitors, when watching smart TVs as visitors in friends' or family's houses, most of the participants used owners' accounts and asked the owners to control TVs. Only two participants had experience logging in to their accounts for convenience reasons.

*Concerns.* All 10 participants indicated limited privacy concerns about using smart TVs in acquaintance's residences. They felt no difference concerning service providers collecting their data through acquaintances' and their own smart TVs and also had the same level of concern about system security on both types of smart TVs. Additionally, no participants expressed concerns about smart TV owners accessing their digital footprints left on the devices.

### 5.2.2 Smart TVs in Public or Shared Spaces.

*Device Use Behavior.* 15 participants discussed their experience interacting with smart TVs in Airbnbs and hotels. Usually, smart TVs in Airbnbs and hotels had default accounts logged in to available streaming services, which many of the participants used during their stay. Two participants also accidentally used accounts that belonged to former guests who forgot to log out. However, several participants would still log in to their personal accounts on public smart TVs. For example, P21 wanted to get access to paid services, for which they could only achieve so by using their personal accounts :

*You can't really access a lot of streaming services without having your own account. And so, if I want to see specific content on a streaming website, I would have to log into my account in order to access it. (P21)*

P11, although usually used pre-logged-in services, logged in to their personal account once for a long stay in Airbnb for convenience's sake.

*Concerns.* Compared to using personal smart TVs or acquaintances' smart TVs, participants expressed an increased level of concern about using smart TVs in public or shared spaces such as Airbnbs and hotels, concerned about both system security and other people accessing their information physically.

Three participants spoke about their uncertainty about the security of public smart TVs. For example, P13 expressed their worries about potential malicious software installed in public smart TVs:

*All these OSs can be...You can sideload on basically everything, all the major ones, and so you never know if they have weird packet sniffers on there or something like that. (P13)*

P12, meanwhile, considered public Internet used by smart TVs in hotels introducing potential security and safety risks.

Participants reported mixed levels of concern about others accessing their information through public smart TVs, depending on which account they used. In terms of using default accounts, participants expressed only limited concern about their information being accessed. They found the information disclosed was not linked to their personal identity and therefore felt fine about any digital footprint being accessed by other guests, staff, or hosts. For example, P16 considered their action through a default account to only “*alter the algorithms that are trying to target ads or content.*” However, when using personal accounts, many participants highlighted their aggregated concerns about other people using their accounts and accessing account information if they did not log off.

*Now they're seeing my YouTube history because I've logged in and I've left [...] I find it to be annoying, first of all. I don't want to know what they're watching and vice versa. (P3)*  
*I do care that there's the potential for [other visitors] to either try to gain control of my account or to ultimately find information like my payment or something like that. (P16)*

**Protective Measures.** In response to the risks of public smart TVs, six participants always avoided using their own account or giving out their personal information:

*I've tended to avoid entering my own information to log in specifically. If I have to do that [to watch the smart TV] I would just use my laptop and my phone. (P1)*

For those who log in to their personal account for convenience, remembering to log off before leaving was the most frequently reported strategy to protect themselves from privacy and security risks. However, this strategy was not always helpful. Some participants reported their frequent uncertainty about whether they truly logged off their account after they left. P20 even reported an incident of themselves forgetting to log off during an Airbnb stay:

*There was a new profile on our Netflix account. I don't understand. It is a new profile, with a name of somebody that we have no idea who it was. [...] I just deleted the account and moved on [...] Maybe we didn't log off it, and someone came in and just opened our account. (P20)*

Moreover, logging off accounts was not enough to prevent information disclosure. For example, P7 pointed out that it could not “*totally remove my footprint on that TV*”. P11 underlined features like “*password manager*”, which, similar to what a computer would do, could leave authentication information on the TV even after logging off the account.

## 6 Indications and Recommendations

The findings presented in section 4 reveal that most participants held negative attitudes towards the current privacy practices of their smart TVs, irrespective of the data collector. A recurring theme throughout our interviews was the perceived lack of choice. Although previous works have established that users often trade privacy for convenience in smart home devices [24, 49, 64, 75], we found this trade-off to be particularly reluctant and resignation-based in the case of smart TVs, with participants expressing a sense of powerlessness, which highlights a significant deficiency in the privacy practices. This is exacerbated by the lack of transparent communication, misaligned

expectations, and the limited control caused by inaccessible settings menu, which further explains why users rarely utilize smart home device settings [28, 53, 72].

Conversely, as discussed in section 5, participants exhibited fewer privacy concerns when using smart TVs around people they know. This relaxed attitude was strongly correlated with the more private settings in which these interactions occur and the trust they place in friends and family, which is similar to how user trust in providers is a key factor for smart home IoT adoption as found in [27]. Notably, when using devices that belong to acquaintances, participants reported no concerns about smart TV owners accessing their digital footprint. We found this result different from reports on visitors' privacy perception of other smart home IoT devices when visiting acquaintances' homes, where they tend to have increased needs of control and privacy concerns [6, 41]. This discrepancy may come from the different types of data collected by smart TVs, which barely include physical data that people are worried about the most (e.g., video and audio recordings) [13, 36, 64]. In fact, unlike other smart home devices where unconsented data collection from visitors is the primary concern [5, 6, 41, 71], smart TVs tend to expose more information about the owner. As discussed in Section 5.1.1, owners may feel uncomfortable when visitors view their watch/search history or personalized recommendations albeit sometimes accidentally. In public settings, such as hotels or Airbnbs, participants were more cautious about safeguarding their information, often avoiding logging into their personal accounts. However, they also experienced inconvenience due to the inability to access their subscription services when restricted to using default accounts on public smart TVs due to these precautions.

Based on our findings, we make the following recommendations to product designers, policy-makers, and researchers. We hope our suggestions can help with enhancing privacy protection and improving user experience.

## 6.1 Privacy Oriented Design

*Privacy Control.* In contrast to prior results, privacy and security were not listed as major considerations during smart TV purchases by our participants [17]. Yet our findings suggest that users, usually less motivated to spend efforts managing their privacy settings when using smart TVs, are frustrated by current privacy controls, which are complex and usually located deeply in settings (Section 4.3). Previous research studying other smart devices and online platforms already highlighted users' limited awareness of privacy management [28, 37, 72] and poor usability of privacy settings [20]. We consider this problem more crucial in the case of smart TVs due to the use case. Similar to the situation with streaming service settings [59], users have limited ability to toggle privacy settings for both the device itself and the third-party apps when watching content on their smart TVs without stopping the video and taking long paths to the settings interface. Building upon suggestions from previous research on a more transparent and meaningful privacy control in smart home IoT devices [20, 28, 54], we advocate that privacy controls should be simple, clear, and appear in the settings as an obvious, standalone category that users can see immediately after they enter the settings menu.

Moreover, as reported in Section 4, many participants drew parallels between their smart TVs and other IoT devices like smartphones and computers due to the similarity in functionalities, and therefore hoped to have the same level of privacy control on their smart TVs. Granted, devices like smartphones have been through more generations and have more developed privacy practice standards. However, if users can control their data in a better way on other devices, they should have the same level of control on their smart TVs as well. Recommended mechanisms include, but are not limited to, notifying users the first time any type of personal data is accessed by a particular app, and regular reminders about their data being accessed by which apps.

*Protection against Shoulder Surfing.* Like many devices with screens, bystanders could easily shoulder surf information, from passwords to watching content, presented on the smart TV interface (Section 5.1.1). Additionally, unlike other IoT devices in a multi-resident household, in which one resident – the primary user – often has the most power of control over them [8, 21, 33, 72], multi-resident-obtained smart TVs usually have no prominent primary users. Having equal chances to control the TV makes it more likely for one user to see information of an account that is previously logged in by other users. When sharing smart TVs with visitors, the owner-dominant control model described by participants, also seen when residents share other IoT devices with visitors [45], may prevent unwanted glance to a limited extent. However, the social boundary of sharing smart TVs with bystanders is more blurred overall, compared to that of sharing other IoT devices. For instance, visitors asking owners for permission, which is a common social manner when sharing smart speakers [45], was not reported by our participants.

Participants reported few personal privacy concerns but expressed a feeling about boundary intrusion on bystanders shoulder surfing their watching information (Section 5.1.3). These findings echo previous research on shoulder surfing on mobile devices, that users usually perceive shoulder surfing as harmless but hold negative feelings towards it for user experience and social relationship reasons [16]. Protection against content-based shoulder surfing, which is much more common than authentication-based one in the wild [16, 19, 60], usually only relies on users' physical actions rather than technical defensive mechanisms [16, 18, 19]. However, protective strategies for mobile phones like shading and moving away the device are unfeasible for smart TVs due to the large screen size.

Therefore, we suggest developing protective mechanisms for both authentication- and content-based shoulder surfing on smart TVs. For authentication-based shoulder surfing, we suggest avoiding situations of inputting passwords using remote controls and instead using smartphone-based authentication and device-free authentication, which has already been considered and deployed for smart home IoT (e.g., [56, 61]). For content-based shoulder surfing, we suggest the deployment of a “guest mode” for streaming services, which could hide any watching information such as recommendations and watching history on the TV interface temporarily.

*Protection Mechanism for Public Devices.* Confirming what is reported by prior works, that visitors care about their relationship with the device owner and familiarity with the environment when considering their privacy [14, 69, 71], participants held different opinions about using acquaintances' smart TVs and those in public or shared spaces. One major concern among participants when using public smart TVs, such as those found in hotels or Airbnbs, was the risk of leaking their account information by logging into their personal accounts. While users might log in to access subscription-based content, they could forget to log out when they leave, leading to incidents like the one experienced by P20 (Section 5.2.2).

Considering the high utility of smart TVs in public spaces [40, 42], it is important to implement privacy protective mechanisms for public smart TVs. Public smart TVs could equip a private browsing mode, which many users adopted to protect their digital footprints from both system security risks and exposure to other users when using public computers [3]. Moreover, to mitigate the risks of forgetting to log out personal accounts, an “auto log-out” mechanism could be implemented. This feature would prompt users to specify how long they want their account to remain logged in, with options ranging from a set number of days or months to a permanent login for use on their own smart TV.

## 6.2 Regulations and Industry Standards

While smart TVs are designed to be increasingly versatile, requiring access to a broader range of data, the corresponding policies have failed to keep pace. The lack of standardized regulations has led to great divergence in privacy policies among different manufacturers [55]. Although the European Union's General Data Privacy Regulation (GDPR) and the California Consumer Privacy Act (CCPA) place significant emphasis on these issues, it remains unclear how well smart TV privacy practices comply with these regulations. In fact, specific smart TV brands have been shown to violate GDPR in a previous study, as well as leaking Personally Identifiable Information (PII) to third parties [2]. Moreover, Tileria et al. found that TV versions of apps provided by the same service providers are significantly less affected by the implementation of applicable policies than their mobile counterparts [66]. This discrepancy is clearly unreasonable and highlights the need for more comprehensive policy enforcement.

Refined regulation of privacy policies is also essential. Echoing the study by Orlowski et al. [53], our findings reveal significant dissatisfaction with how smart TV privacy practices are communicated to users. Participants reported that they never read privacy policies, or, even if they attempted to, they found the language too complex to understand (Section 4.3). Privacy policies should not only serve as legal documents but also as an effective tool for informing users about data collection and usage. Furthermore, User consent cannot be considered valid if users do not fully comprehend what they are agreeing to. Participants strongly advocated for the inclusion of a section where information is presented in an easily understandable format. For example, using lists or tables to clearly outline the types of data collected and how that data is used was highly recommended. Additionally, users should have continuous access to this information, rather than only encountering it when they initially agree to the privacy policy upon first using the service.

## 6.3 Future Research Directions

Researchers often serve as the significant role of a bridge that connects policymakers and users, helping with both policy-making and enforcement and user privacy protection. Our results raise important research questions and we believe addressing these research gaps would not only enhance our understanding of user privacy needs but also contribute to the development of more effective and user-friendly privacy solutions.

*User Expectations.* Although we briefly touched on potential privacy control designs preferred by participants, we did not explore user expectations in depth. Future studies should investigate user preferences regarding various privacy control designs more thoroughly, examining how these preferences may differ across different devices, such as smartphones or computers. Understanding these preferences could guide the development of more user-centric privacy controls. Additionally, user expectations regarding privacy regulation policies require further attention. While our participants lacked detailed knowledge of privacy-related policies, they did express ideas about how service providers should be regulated. Further exploration of this topic could help align policies with user expectations, thereby enhancing privacy protection.

*Policy enforcement.* Policies lose their value significantly if they are not enforced. Researchers could measure the policy compliance of smart TV software to enhance policy enforcement. Investigating policy compliance could provide insights into the effectiveness of existing privacy regulations. Additionally, the tools developed for measuring compliance would be invaluable for longitudinal studies, providing a means to track changes over time and assess the long-term impact of privacy regulations.

## 7 Conclusion

Through semi-structured interviews with 22 smart TV users, our study explores user privacy concerns related to smart TVs from both provider and social perspectives. While users value the convenience and easy access to content, they express concerns about data collection by manufacturers and third-party apps, which is further exacerbated by unclear and complex privacy settings. The shared use of smart TVs also raises privacy challenges, particularly with bystanders or in public spaces, though perceived trust in acquaintances and the non-sensitive nature of shared data often reduce these concerns in private settings. Our findings highlight the need for clearer privacy communication, better controls, and a deeper understanding of social contexts to enhance privacy protection and improve smart TV user experience.

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### A Supplementary Information of Participants

| Participant | Occupation                                     | Use Smart TVs for                                                                                  | Use smart TVs with                                        |
|-------------|------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| P1          | Professional, scientific or technical services | Watch online videos                                                                                | By yourself, With your family memebers, With your friends |
| P2          | Other services (except public administration)  | Watch online videos, Watch livestreams, Play games                                                 | By yourself, With your friends                            |
| P3          | Information                                    | Watch traditional broadcast TV, Watch on-line videos, Watch livestreams                            | By yourself, With your family memebers, With your friends |
| P4          | Finance or insurance                           | Watch traditional broadcast TV, Watch on-line videos, Watch livestreams, Listen to music / podcast | By yourself                                               |
| P5          | Educational services                           | Watch traditional broadcast TV                                                                     | With your family memebers                                 |
| P6          | Information                                    | Watch online videos, Watch livestreams                                                             | With your friends                                         |
| P7          | Educational services                           | Watch online videos, Use as regular monitors                                                       | By yourself                                               |
| P8          | Professional, scientific or technical services | Watch traditional broadcast TV, Watch on-line videos                                               | By yourself                                               |
| P9          | Professional, scientific or technical services | Watch online videos, Watch livestreams, Listen to music / podcast                                  | With your family members                                  |
| P10         | Information                                    | Watch online videos, Play games                                                                    | By yourself                                               |
| P11         | Professional, scientific or technical services | Watch traditional broadcast TV                                                                     | With your family members                                  |
| P12         | Professional, scientific or technical services | Watch traditional broadcast TV, Watch on-line videos, Watch livestreams, Play games                | By yourself, With your family members                     |
| P13         | Health care or social assistance               | Watch online videos, Watch livestreams, Listen to music / podcast                                  | By yourself, With your family members                     |
| P14         | Information                                    | Watch traditional broadcast TV, Watch on-line videos                                               | By yourself, With your family members                     |
| P15         | Unclassified establishments                    | Watch online videos, Watch livestreams                                                             | By yourself, With your family members                     |
| P16         | Educational services                           | Watch traditional broadcast TV, Watch on-line videos, Watch livestreams                            | By yourself, With your family members                     |
| P17         | Professional, scientific or technical services | Watch traditional broadcast TV, Watch on-line videos                                               | With your family members                                  |
| P18         | Educational services                           | Watch traditional broadcast TV, Watch on-line videos                                               | By yourself, With your family members                     |
| P19         | Finance or insurance                           | Watch traditional broadcast TV, Watch on-line videos                                               | By yourself, With your family members                     |
| P20         | Other services (except public administration)  | Watch traditional broadcast TV, Watch on-line videos                                               | By yourself, With your family members, With your friends  |
| P21         | Educational services                           | Watch online videos, Watch livestreams                                                             | By yourself, With your friends                            |
| P22         | Educational services                           | Watch traditional broadcast TV, Watch on-line videos, Listen to music / podcast, Play games        | By yourself, With your family members                     |

Table 2. Supplementary Demographic Information of Participants

## B Surveys and Interview Scripts

### B.1 Screening Survey Questions

- (1) Have you used any smart TVs from the following brands or with the following operation systems?
- (2) On average, how many days per week do you use Smart TVs?
- (3) On average, how many days per month do you use Smart TVs?
- (4) Do you own any smart TVs from the following brands or with the following operation systems?
- (5) How many smart TVs do you own in total?
- (6) What services or Apps do you use most frequently on smart TVs?
- (7) What activities do you usually use Smart TVs for?
- (8) What is your age?
- (9) What gender do you identify with?
- (10) Do you live in the U.S.?
- (11) What is your highest level of education attained?
- (12) Which of the following industries most closely matches the one in which you are employed?

### B.2 Pre-Interview Survey Questions

- (1) At which of the following places do you usually watch smart TVs?
- (2) Who do you usually watch smart TV with?
- (3) What size is your smart TV?
- (4) Where is your smart TV placed?
- (5) In what ways do you usually control your smart TV(s)?

### B.3 Interview Scripts

#### B.3.1 General Usage.

- (1) How would you describe smart TVs to someone who has never used one before?
- (2) Why did you purchase a smart TV?
  - Why did you choose a smart TV over a traditional TV?
  - Are there any factors that affect your decision to buy a [brand] smart TV over other brands?
- (3) What is your primary purpose for using your smart TV?
  - What type of videos or livestreams do you usually watch on your smart TV?
  - Which apps or third-party services do you use most frequently?
- (4) Now, let's talk about how you use your Smart TV.
  - When you want to watch a video or livestream, do you prefer to search for specific content, or browse the recommendation list, or it depends? Please describe the series of actions you take when you want to watch something on a smart TV.
  - Have you ever done anything using your smart TV beyond watching content? If so, please describe the process to me. If not, is there a specific reason for you to not do so? How would you do it instead?
- (5) Are there other people in your residence that are also users of your smart TV?
  - Does everyone have equal access to the smart TV?
  - Tell me more about who uses which accounts on your smart TV.
- (6) You also mentioned in the survey that you watch your smart TV with [ others ].
  - Who usually controls your smart TV?
  - Do they have access to your smart TV? If so, do you share accounts with them when they watch your TV?

### B.3.2 *Data Collection and Processing.*

- (1) What do you think your smart TV knows about you?
  - Which of them do you consider as personal or private?
  - What do you think happens to these sets of information after they are collected?
  - Who do you think has access to these sets of information?
- (2) What kinds of information do you think your smart TV manufacturer has access to?
  - What do you think your smart TV manufacturer would do with your personal information?
  - How comfortable do you feel about your smart TV manufacturer accessing your personal information?
- (3) What kinds of information do you think the third-party apps you use has access to?
  - What do you think the third-party apps would do with your personal information?
  - How comfortable do you feel about the third-party apps accessing your personal information?
- (4) How comfortable do you feel about your watching and searching history accessed or used by other entities?
  - By which entities do you not want your watching and searching history accessed or used?
  - What do you think will happen to your watching and searching history?
- (5) How satisfied are you with the current level of control you have over your personal information on your smart TV?
- (6) In what ways does your smart TV inform you about or allow you to have control over your personal information? Could you give me some specific examples?
- (7) Smart TV users can control their smart TV(s) through remote controls, their phones, or their smart speakers, do you think any of them are more secure than others? Why?

### B.3.3 *Account Sharing and Shoulder Surfing.*

- (1) What kinds of personal information on the TV do you not want others around you to access?
- (2) Do you think others, for example, your roommates, friends, or even people walking past your window, can see your personal information on your smart TV? These may include information related to your accounts, recommendations, etc.
  - Have you ever tried to prevent other people from seeing your personal information when they watching your TV with you or passing by?
- (3) Overall, does others' presence affect how you interact with your smart TV? If so, how?
- (4) You mentioned that you share your accounts with [others] when they watch your smart TV, what do you think are the pros and cons of sharing your account with them?
  - How comfortable do you feel about others seeing the your information when using your account?
  - Have you ever tried to prevent other people from getting your personal information when they use your account?
- (5) You mentioned that you also use [other]'s account on your smart TV, what do you think are the pros and cons of doing so?
  - Do you consider your activities on others' accounts might disclose your personal information? How comfortable are you?
  - Have you ever tried to prevent your personal information from being disclosed when using others' account?

### B.3.4 *Public Devices Usage.*

- (1) What do you concern about when using smart TVs in other places other than your house (e.g. partner's or friends' place / airbnb and hotel)

- Do you think your activities on their smart TV might disclose your personal information?
  - Do you use other’s accounts when watching their smart TV?
  - Have you ever logged into your own account on their smart TV?
  - How comfortable do you feel about the smart TV owner accessing your personal information through their TV?
  - Have you ever tried to prevent your personal information from being disclosed on other’s smart TV?
- (2) Compared to your own smart TV, do you think using a smart TV that does not belong to you exposes you to more or less privacy and security risks?

*B.3.5 Security and Privacy Awareness.*

- (1) Do you know of any security attacks on smart TVs?
- (2) How much do you think your usage of smart TV is affected by the attacks?
- (3) Would the development of security attacks cause you to switch back to a traditional TV?
- (4) Do you consider yourself someone with a high privacy awareness? You can rate yourself using the scale of high, medium, or low.
- (5) Do you have any privacy-related concerns about technology in general?

**C Codebook**

| General Code            | Sub-Code                            | Description                                                                       |
|-------------------------|-------------------------------------|-----------------------------------------------------------------------------------|
| Smart TV concept        |                                     | How the participants described smart TVs.                                         |
| Purchase considerations |                                     |                                                                                   |
|                         | Reason for buying                   | Why the participant bought their smart TVs.                                       |
|                         | Reason for not buying               | Why the participant didn’t buy a smart TV or their concerns of buying a smart TV. |
| Purposes and habits     |                                     |                                                                                   |
|                         | Purpose of using                    | What the participant use their smart TVs for.                                     |
|                         | Watching habit                      | How the participant watch content. e.g. browsing, searching, etc.                 |
|                         | Activities besides watching content | What the participant use their smart TVs for besides watching content.            |

Table 3. Codes under the “General Usage” category

| General Code                   | Sub-Code                      | Description                                                                                                                          |
|--------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Private info definition        |                               | What the participant considers as private or personal.                                                                               |
| General info collected         |                               | Information the participant thinks is collected by their smart TVs.                                                                  |
| Manufactures                   |                               |                                                                                                                                      |
|                                | Manufactures collect          | What information the participant thinks the TV manufacturer collects.                                                                |
|                                | Manufactures process          | What the participant thinks the TV manufacturer does with their information.                                                         |
|                                | Manufactures comfort level    | How comfortable the participant feels about how the TV manufacturer uses their information.                                          |
| Third-parties                  |                               |                                                                                                                                      |
|                                | Third-parties collect         | What information the participant thinks the third-party apps collect.                                                                |
|                                | Third-parties process         | What the participant thinks the third-party apps do with their information.                                                          |
|                                | Third-parties comfort level   | How comfortable the participant feels about how the third-party apps use their information.                                          |
| Manufactures vs. third-parties |                               | How the participant may feel differently about manufacturers and third-parties or how manufacturers and third-parties work together. |
| Other collectors               |                               |                                                                                                                                      |
|                                | Collector                     | Who the participant thinks collects their information through their smart TVs besides manufacturers and third-parties.               |
|                                | Information collected         | What information the participant thinks is collected through their smart TVs besides manufacturers and third-parties.                |
| Watch/search history           |                               | The participant's comment on watch and search history specifically.                                                                  |
| Privacy control and inform     |                               |                                                                                                                                      |
|                                | Control perception            | Whether the participant finds the level of control they have over their information on their smart TVs satisfactory and why.         |
|                                | Control suggestion            | The participant's suggestion about how to improve the control over their information on their smart TVs.                             |
|                                | Inform perception             | Whether the participant finds how their smart TVs inform them about data collection and processing satisfactory and why.             |
|                                | Inform suggestion             | The participant's suggestion about how their smart TVs can inform them about data collection and processing.                         |
| Ways of control                |                               |                                                                                                                                      |
|                                | Secure ways of control        | Which ways of control participant thinks is the most secure.                                                                         |
|                                | Perception of ways of control | What concerns the participant has about controlling their smart TVs via which ways.                                                  |

Table 4. Codes under the “Data Collection and Processing” category

| General Code                                       | Sub-Code                                    | Description                                                                                                  |
|----------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Account sharing                                    |                                             |                                                                                                              |
|                                                    | Control when sharing                        | Who usually controls the participant's smart TVs when shared by others.                                      |
|                                                    | Account sharing habit                       | How participant shares accounts on their smart TVs.                                                          |
|                                                    | Pros of sharing accounts                    | What the participant thinks are the benefits of sharing accounts.                                            |
|                                                    | Cons of sharing accounts                    | The participant's concern about sharing accounts.                                                            |
|                                                    | Perceptions of using others' accounts       | The participant's concern about using others' accounts.                                                      |
|                                                    | Perceptions of sharing accounts with others | The participant's concern about sharing their own accounts with others.                                      |
| Co-watchers                                        |                                             |                                                                                                              |
|                                                    | Watching together see info                  | What information the participant thinks the people watching smart TV with them can see.                      |
|                                                    | Watching together see info comfort level    | How comfortable the participant feels about the people watching smart TV with them seeing their information. |
|                                                    | Passing by see info                         | What information the participant thinks the people passing by can see.                                       |
|                                                    | Passing by see info comfort level           | How comfortable the participant feels about the people passing by seeing their information.                  |
|                                                    | Affects of others' presence                 | How the presence of others affects how the participant uses their smart TVs.                                 |
| Attempts to prevent others from seeing info        |                                             | What the participant does to prevent their information from leaking on a personal smart TV.                  |
| Information disclosure in acquaintance's home      |                                             | What information the participant thinks is disclosed when using acquaintances' smart TVs.                    |
| Account use and perception in acquaintances' place |                                             | How participant share accounts at acquaintances' place and how comfortable the participant feels about it.   |

Table 5. Codes under the "Private Space Sharing" category

| General Code                        | Sub-Code                           | Description                                                                                                               |
|-------------------------------------|------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Reason for using or not using       |                                    | Why the participant does or does not want to use smart TVs in public spaces.                                              |
| Information disclosure              |                                    | What information the participant thinks would be disclosed by using a public smart TV.                                    |
| Account using and perception        |                                    |                                                                                                                           |
|                                     | Account using                      | What account the participant uses when using a public smart TV and whether they log out of their own account after use.   |
|                                     | Personal account log-in perception | Participant’s concerns about using their personal account on a public smart TV.                                           |
|                                     | TV owner access info comfort level | How comfortable the participant feels about the public smart TV owner accessing their information on the public smart TV. |
| Attempts to prevent info disclosure |                                    | What the participant does to prevent their information from leaking on a public smart TV.                                 |
| TV owner access info comfort level  |                                    | How comfortable the participant feels about the public smart TV owner accessing their information on the public smart TV. |

Table 6. Codes under the “Public Space Sharing” category

| General Code             | Sub-Code                      | Description                                                                                                               |
|--------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Attack perception        |                               |                                                                                                                           |
|                          | Security attacks              | Participant’s knowledge about security attacks on smart TVs.                                                              |
|                          | Switch back to traditional TV | Whether the participant would switch back to traditional TVs because of the development of security attacks on smart TVs. |
| Privacy awareness rating |                               | Participant’s self-rated privacy awareness.                                                                               |
| General tech concerns    |                               | Participant’s concerns about technology in general.                                                                       |

Table 7. Codes under the “Security and Privacy” category

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